



揭榜挂帅第一次线上汇报 ——智能算法系统

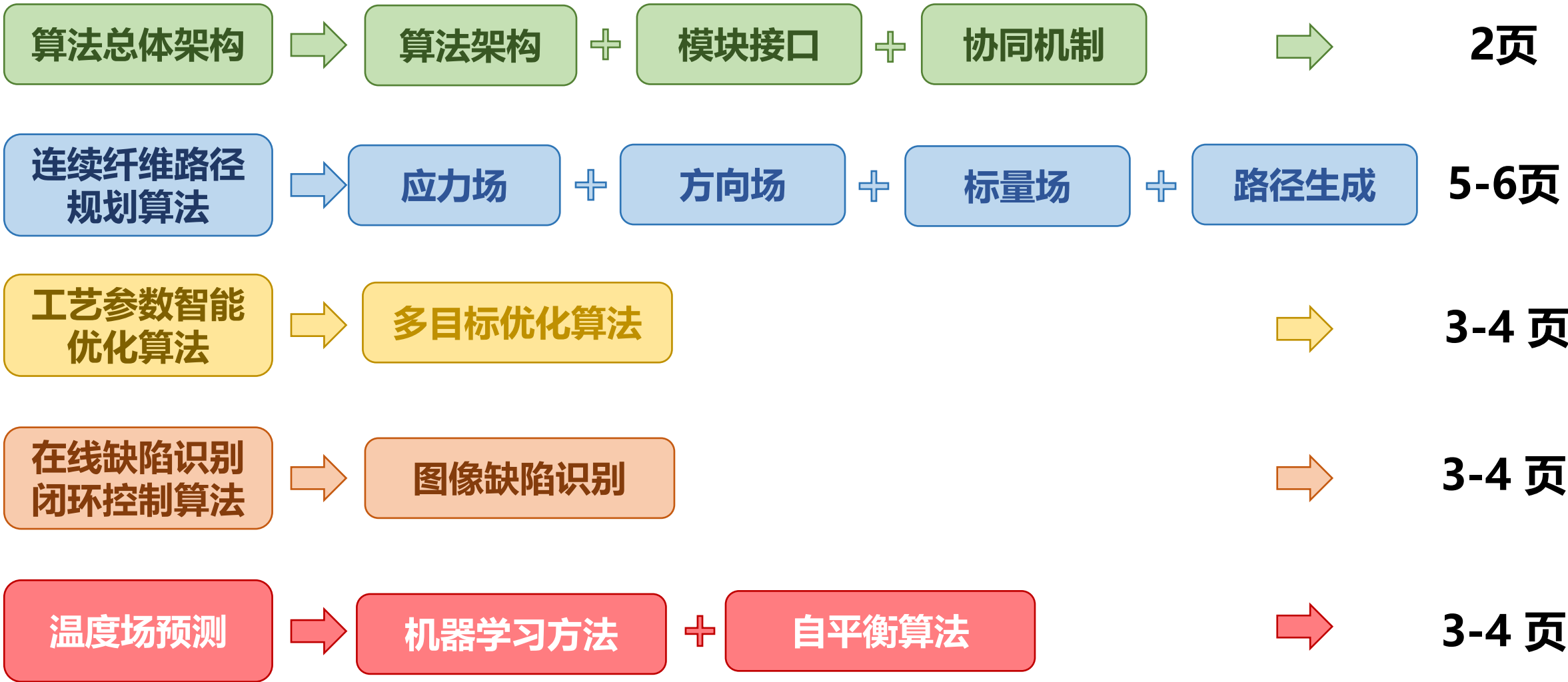


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2026年 7月14日



框架整理：智能算法模块文本思路

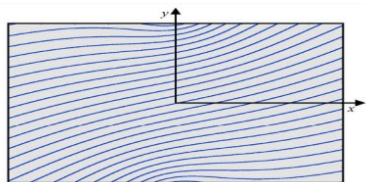
文本框架



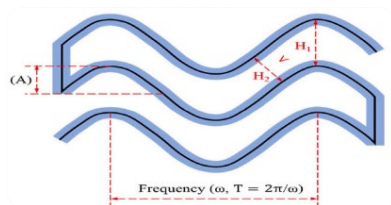
5.2 连续纤维路径规划算法—调研与认知

1. 构造函数法

方法特性：建立线性、二次、三次、三角函数、B样条曲线等路径函数



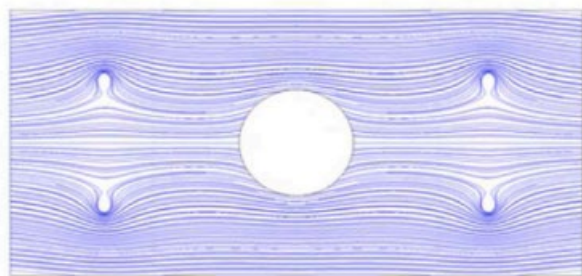
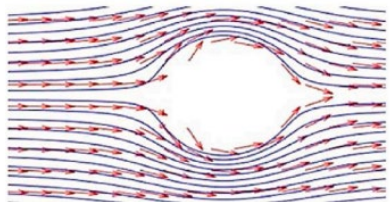
流场函数



正弦函数

2. 流体流线法

方法特性：沿流体流线方向设计非交叉曲线纤维轨迹



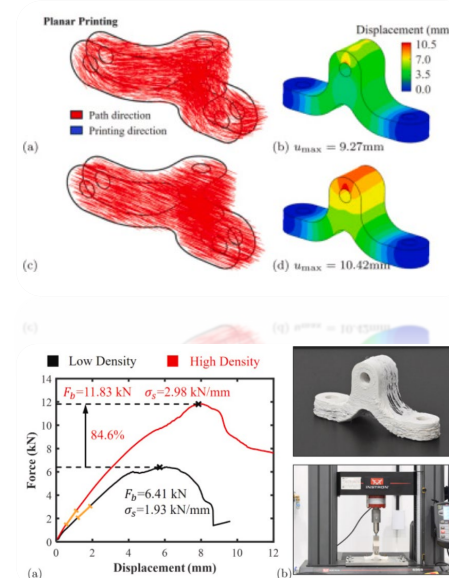
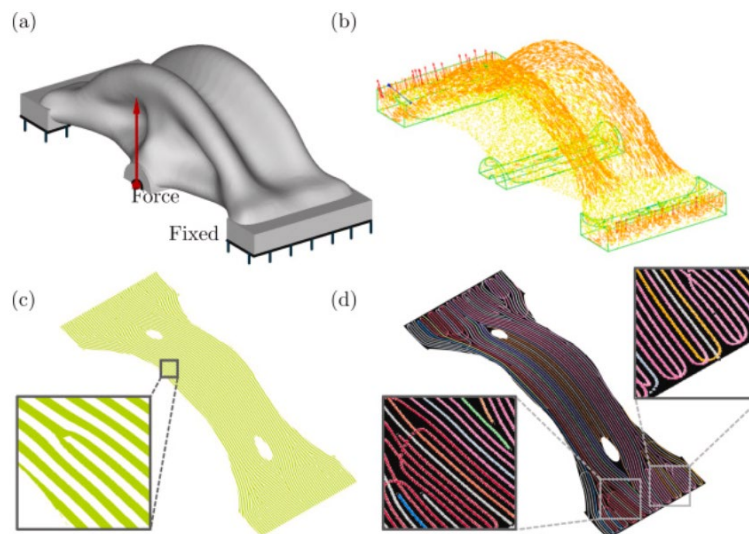
3. 应力法 (10篇文献)

(1) 隐式规划法：(不同尺寸的映射)

有限元分析->切片分析应力->每片投影->依据标量场生成

(2) 显式规划法：

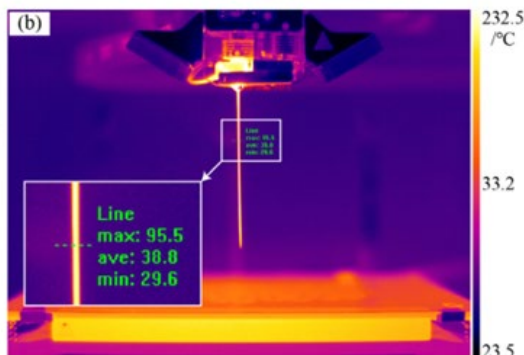
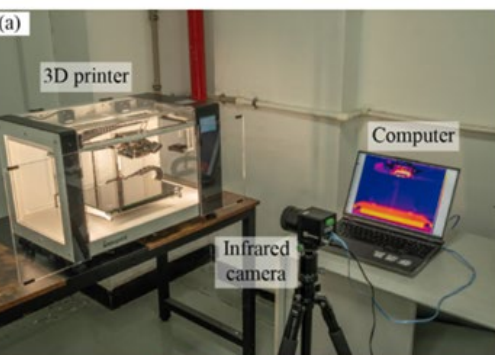
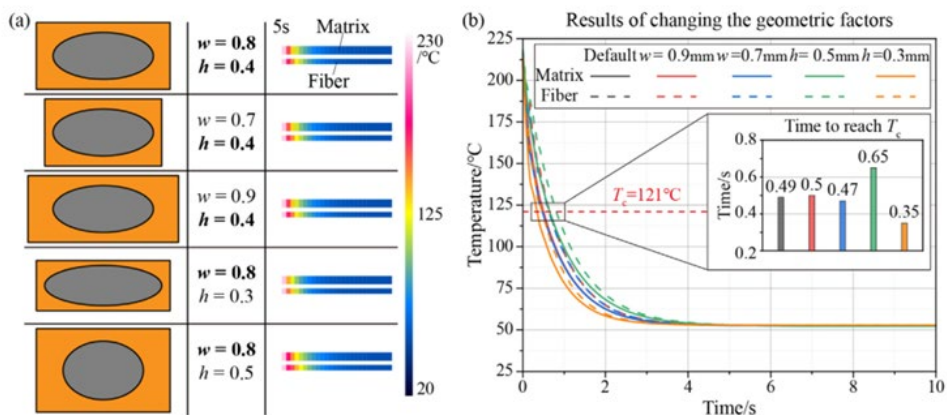
有限元分析->切片分析应力->直接根据曲面流线生成轨迹



5.3 工艺参数优化方法—调研与认知

1. 实验验证法

方法特性：设定一连串参数，打印后，测试打印件强度



2. 优化算法 (仿真作为数据)

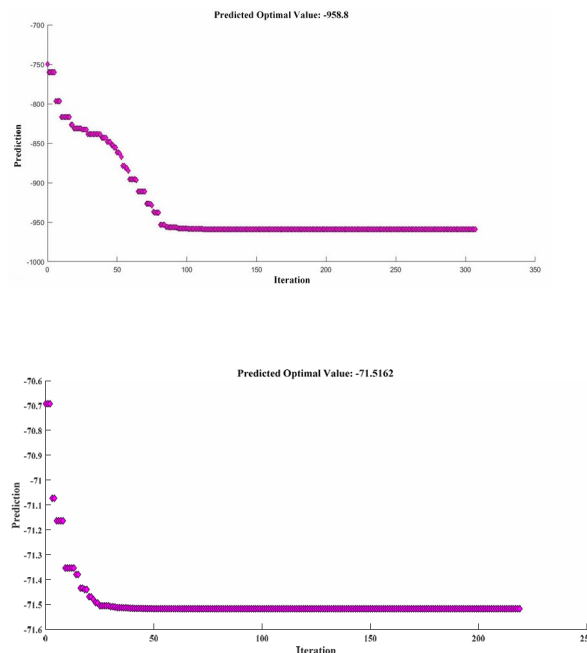
(1) 机器学习算法优化

(2) 常见优化算法：模拟退火，遗传等

Table 2. Experimental data.

#	Printing Speed (mm min ⁻¹)	Printing Space (mm)	Printing Thickness (mm)	Curing Temperature ($^\circ\text{C}$)	Curing Pressure (MPa)	Flexural Strength (MPa)	Flexural Modulus (GPa)
1	200	1.0	0.25	150	-0.02	660.8699	57.5127
2	200	1.1	0.30	160	-0.04	717.4586	61.3590
3	200	1.2	0.35	170	-0.06	801.9369	65.4731
4	200	1.3	0.40	180	-0.08	767.1566	58.6598
5	200	1.4	0.45	190	-0.10	666.3564	54.9489
6	500	1.1	0.25	170	-0.08	861.6027	71.6292
7	500	1.2	0.30	180	-0.10	712.3392	58.8811
8	500	1.3	0.35	190	-0.02	695.9950	65.8205
9	500	1.4	0.40	150	-0.04	740.2541	57.6429
10	500	1.0	0.45	160	-0.06	842.4707	67.3847
11	800	1.2	0.25	190	-0.04	616.9910	44.3187
12	800	1.3	0.30	150	-0.06	636.2180	48.6564
13	800	1.4	0.35	160	-0.08	792.8516	63.6566
14	800	1.0	0.40	170	-0.10	697.4975	57.5773
15	800	1.1	0.45	180	-0.02	674.6801	53.1469
16	1100	1.3	0.25	160	-0.10	682.3967	51.4495
17	1100	1.4	0.30	170	-0.02	636.5848	48.9007
18	1100	1.0	0.35	180	-0.04	679.3136	50.8338
19	1100	1.1	0.40	190	-0.06	783.9741	60.8011
20	1100	1.2	0.45	150	-0.08	817.1150	66.3398
21	1400	1.4	0.25	180	-0.06	778.9613	63.7432
22	1400	1.0	0.30	190	-0.08	721.1591	57.8544
23	1400	1.1	0.35	150	-0.10	702.9702	53.3820
24	1400	1.2	0.40	160	-0.02	707.0509	53.7000
25	1400	1.3	0.45	170	-0.04	745.9622	52.9382
26	200	1.2	0.35	150	-0.10	765.1427	53.63408
27	500	1.2	0.35	150	-0.10	736.4206	53.44402
28	800	1.2	0.35	150	-0.10	746.3091	56.4504
29	1100	1.2	0.35	150	-0.10	682.7215	51.20623
30	1400	1.2	0.35	150	-0.10	634.1206	46.95916
31	800	1.2	0.25	150	-0.10	612.8753	50.9161
32	800	1.2	0.30	150	-0.10	715.8942	56.92083
33	800	1.2	0.40	150	-0.10	737.8536	52.44632
34	800	1.2	0.45	150	-0.10	742.5152	55.96409
35	800	1.0	0.35	150	-0.10	661.6145	49.49135
36	800	1.1	0.35	150	-0.10	683.9867	48.4565
37	800	1.3	0.35	150	-0.10	745.0633	58.17614
38	800	1.4	0.35	150	-0.10	821.5221	64.13514
39	800	1.2	0.35	150	-0.08	826.4611	64.51862
40	800	1.2	0.35	150	-0.06	800.9212	62.79963
41	800	1.2	0.35	150	-0.04	717.5753	62.80635
42	800	1.2	0.35	150	-0.02	648.4216	47.78122
43	800	1.2	0.35	160	-0.10	916.0076	66.69733
44	800	1.2	0.35	170	-0.10	952.8868	71.95371
45	800	1.2	0.35	180	-0.10	754.4841	61.38528
46	800	1.2	0.35	190	-0.10	720.601	51.66744

训练



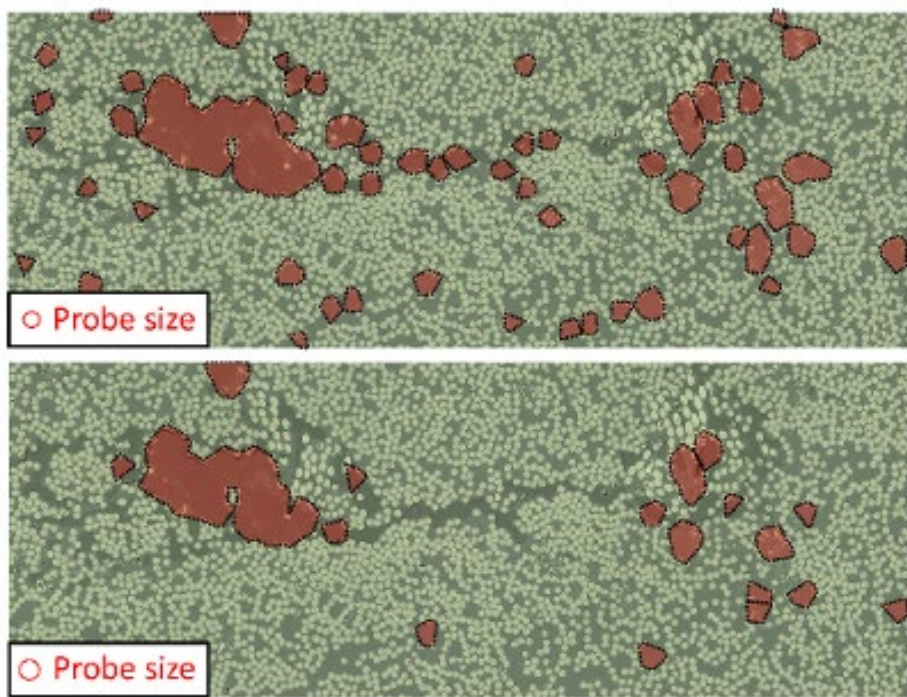
实验数据

预测输出

1. 离线检测法

方法特性：需要将产品移除，甚至破坏来检测缺陷

例：取截面利用YOLO进行检测缺陷：

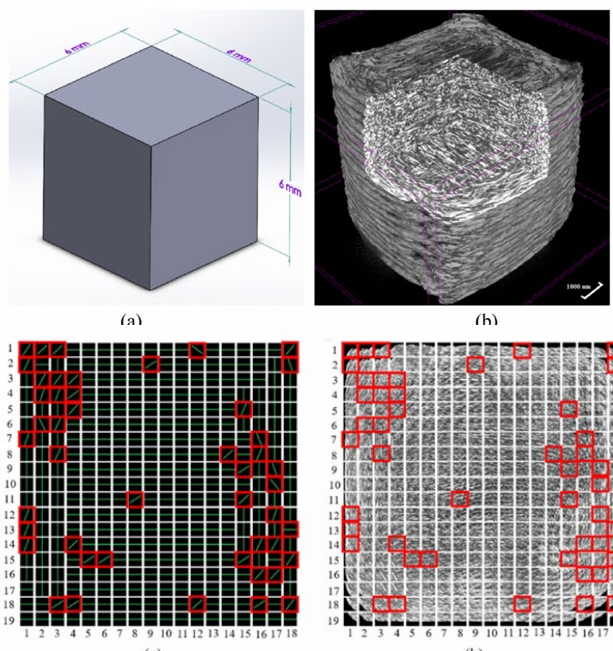


2. 在线检测

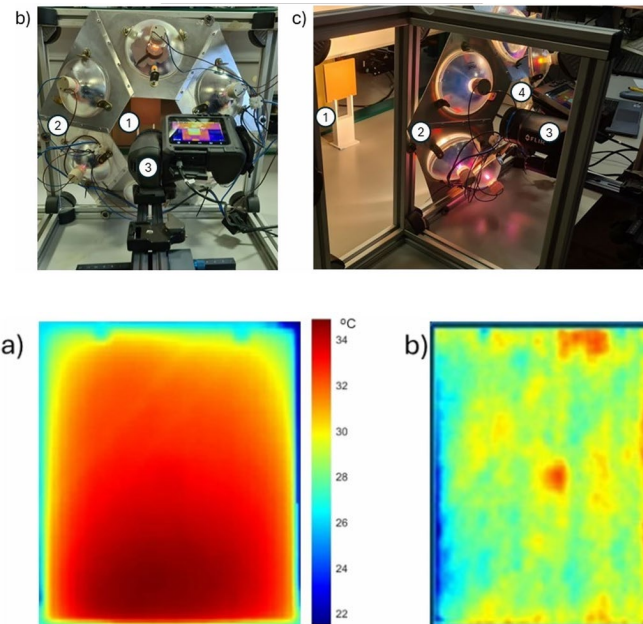
(检测参数的确定, 厚度, 缺陷等常见问题)

(1) 加热检测—对输出图像检测

(2) X光衍射—对输出图像检测



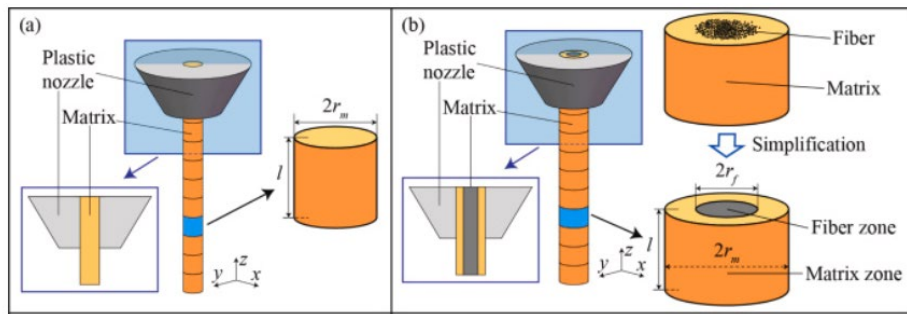
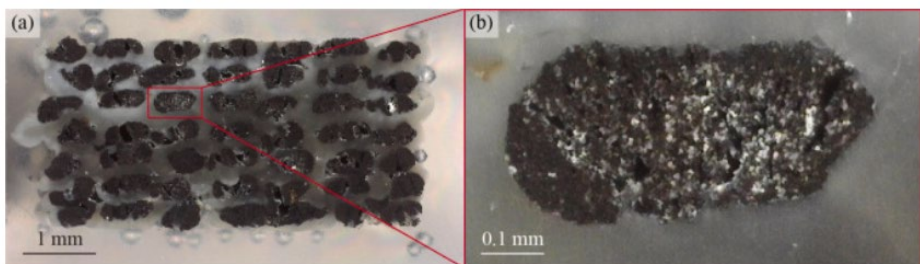
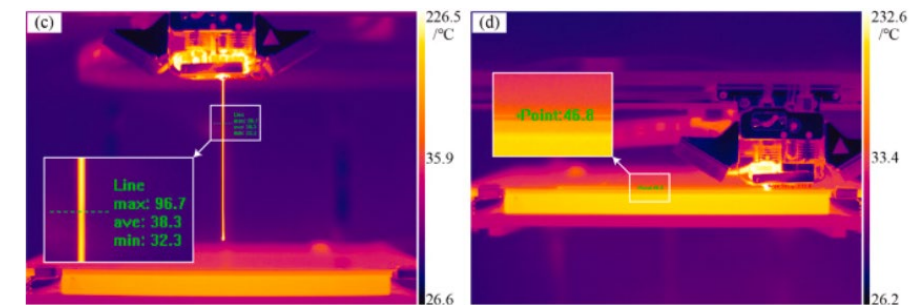
X射线衍射与检测



加热与热辐射检测

1. 实验法

方法特性：设置多组参数，对比试验检测



2. 自主预测

(1) 优化算法+自平衡控制

(2) 函数规划+自平衡控制

